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1 Introduction

1.1 Sets

1.1.1 Equivalence between sets

Lemma 1.1.1

Suppose $f : X \to Y, g : Y \to X$, then there are decompositions

$$X = A \cup \tilde{A}, \quad Y = B \cup \tilde{B}$$

such that $A \cap \tilde{A} = \varnothing, B \cap \tilde{B} = \varnothing$ and $f(A) = B, g\left(\tilde{B}\right) = \tilde{A}$.

Note.
我们从随便一个 A 出发，先让一些简单的条件得到满足，比如 $B = f(A), \tilde{B} = Y \backslash B, \tilde{A} = g(Y \setminus f(E))$，就它们确定地写出来. 剩下 $A \cap \tilde{A} = \varnothing$ 和 $A \cup \tilde{A} = X$ 需要满足. 直接找到特定的目标一开始是困难的，于是我们先把满足 $A \cap \tilde{A} = \varnothing$ 的“好东西”都收集起来，即 $\mathcal{S}$. 再在 $\mathcal{S}$ 中寻找一个满足 $A \cup \tilde{A} = X$ 的候选. “极大元”就是一个看起来非常可疑的对象 (如果它也落在 $\mathcal{S}$，事实上也的确如此).

Proof. Let

$$\mathcal{S} := \{E \subseteq X : E \cap g(Y \setminus f(E)) = \varnothing\}.$$

We define

$$A := \bigcup_{E \in \mathcal{S}} E$$

Let

$$B := f(A), \quad \tilde{B} := Y \setminus f(A), \quad \tilde{A} := g\left(\tilde{B}\right) = g(Y \setminus f(A))$$

It is obvious that $B \cap \tilde{B} = \varnothing, B \cup \tilde{B} = Y$. We should show $A \cap \tilde{A} = \varnothing$, which is equivalent to $A \in \mathcal{S}$, and $A \cup \tilde{A} = X$. First , take any $E \in \mathcal{S}$, we have

$$E \cap g(Y \setminus f(A)) \subseteq E \cap g(Y \setminus f(E)) = \varnothing.$$

Then we concludes that

$$A \cap g(Y \setminus f(A)) = \varnothing \implies A \in \mathcal{S}$$

To show that $A \cup \tilde{A} = X$, we suppose conversely that there is $x \in X$ such that

$$x \notin A \cup \tilde{A}$$

Let $A' := A \cup \{x\}$, then

$$\begin{aligned} A' \cap g(Y \setminus f(A')) &= (A \cup \{x\}) \cap g(Y \setminus (f(A) \cup \{x\})) \\ &\subseteq (A \cup \{x\}) \cap \tilde{A} \\ &= \left(A \cap \tilde{A}\right) \cup \left(\{x\} \cap \tilde{A}\right) \\ &= \varnothing \end{aligned}$$

Thus $A' \in \mathcal{S}$, which is a contradiction to the definition of A. □

Theorem 1.1.2 ▶ Cantro-Bernstein

If $X \supseteq A$, $Y \supseteq B$, and if $X \sim B$, $Y \sim A$, then $X \sim Y$.

Proof. We know that there are injection $f : X \to Y$ and $g : Y \to X$. There are decompositions

$$X = A \cup \tilde{A}, \quad Y = B \cup \tilde{B}$$

such that $A \cap \tilde{A} = \varnothing, B \cap \tilde{B} = \varnothing$ and $f(A) = B$, $g\left(\tilde{B}\right) = \tilde{A}$. Then we define $F : X \to Y$

$$F(x) := \begin{cases} f(x), & x \in A \\ \left(g|_{\tilde{B}}\right)^{-1}(x), & x \in \tilde{A} \end{cases}$$

is a bijection. □

1.1.2 Countable sets and the Cardinality of the Continuum

Example 1.1.3

The following sets are countable:

1. $\{I_\alpha\}_{\alpha \in \Lambda}$, where $I_\alpha \subseteq \mathbb{R}^1$ is an interval and $I_\alpha \cap I_\beta = \varnothing$ whenever $\alpha \neq \beta$.
2. Let $f : \mathbb{R}^1 \to \mathbb{R}^1$ be a monotone function, then

   $$A := \{x : f(x) \text{ is not continuous at } x\}$$

   is a countable set.

3. Let $f : \mathbb{R} \to \mathbb{R}$, define

$$E := \{x \in \mathbb{R} : f(x) \text{ is not continuous on } x, \text{but the right limit } f(x+0) \text{ exists}\}$$

Proof. 1. We take a rational number r_α in each I_α.

2. We take a rational number r_x in $(f(x-0), f(x+0))$ for each x that f is not continuous on it.

□

Theorem 1.1.4

Let A be a infinite set and $|A| = \alpha$. If B is countable, then

$$|A \cup B| = \alpha$$

Proof. Since $B' := B \setminus A$ is countable, and $A \cup B' = A \cup (B \setminus A) = A$, we may assume that $A \cup B = \varnothing$. There exists countable set $A_1 \subseteq A$. Then $A_1 \cup B$ is countable. We have $A_1 \cup B \sim A_1$. There exists bijection $f : A_1 \cup B \to A_1$. Since

$$A = A_1 \cup (A \setminus A_1), \quad A \cup B = (A_1 \cup B) \cup (A \setminus A_1)$$

Define $F : A \cup B \to A$

$$F(x) = \begin{cases} f(x), & x \in A_1 \cup B \\ x, & x \in A \setminus A_1 \end{cases}$$

Then F is a bijection. Thus $|A \cup B| = |A| = \alpha$. □

Theorem 1.1.5

$$A \text{ is a infinite set} \iff \exists A_1 \text{ is a proper subset of } A, s.t. |A_1| = |A|$$

Theorem 1.1.6

$[0,1] = \{x : 0 \leq x \leq 1\}$ is a uncoutable set.

Proof sketch.
试图建立 $(0,1] \sim \mathscr{H}$，其中 $\mathscr{H}$ 表示自然数列全体. 将 $[0,1]$ 上的数按二进制展开 (只采用具有无穷多个 1 的表示). 将展开式从低到高位次, 将两个数码 1 之间

的间隔 (位置差) 组成一个自然数列. 从而建立了 $(0,1] \sim \mathscr{H}$.

Definition 1.1.7

Define

$$c := |(0,1]|$$

called the ***cardinality of the continuum***.

Theorem 1.1.8

Let $\{A_k\}$ be a suquence of sets. If

$$|A_k| = c, \quad \forall k \in \mathbb{N}$$

Then

$$\left|\bigcup_{k=1}^{\infty} A_k\right| = c$$

Proof sketch.
$A_k \sim [k, k+1)$, 故 $\left|\bigcup_{k=1}^{\infty} A_k\right| \leq \left|\coprod_{k=1}^{\infty} A_k\right| \leq |[1,\infty)| = c$. 而 $A_k \subseteq \bigcup_{k=1}^{\infty} A_i \implies \left|\bigcup_{k=1}^{\infty} A_k\right| \geq c$. 因此由 CSB 定理, $\left|\bigcup_{k=1}^{\infty} A_k\right| = c$.

1.2 Arithmetic in [0,infty]

1.3 Metric Space

Lemma 1.3.1

X is a metric space, $x \in X$, B_1, B_2 are twos balls in X. If $x \in B_1 \cap B_2$, then x is the center of an open ball $B \subseteq B_1 \cap B_2$.

Proof. We may assume that B_1, B_2 have different center. Let $y_1 \neq y_2$,

$$B_1 = \{x : d(x, y_1) < r_1\}, \quad B_2 = \{x : d(x, y_2) < r_2\}$$

Take $x \in B_1 \cap B_2$, we claim that there exists $r_0 > 0$ such that $B := \{y : d(x,y) < r_0\} \subseteq B_1 \cap B_2$. Otherwise , for all $r > 0$, there exists a point y_r such that the following two hold at the same time

1. $d(x, y_r) < r$
2. $d(y_r, y_1) \geq r_1$ or $d(y_r, y_2) \geq r_2$.

Thus

$$d(x, y_1) \geq d(y_r, y_1) - d(x, y_r) > r_1 - r, \quad \text{or} \quad d(x, y_2) > r_2 - r$$

The above shows that

$$r > \min\{r_1 - d(x, y_1), r_2 - d(x, y_2)\}, \quad \forall r > 0$$

which is a contradiction. □

1.4 Limits of Set Sequences

Definition 1.4.1 ▸ Limit Inferior and Limit Superior

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. By the ***Limit Inferior of the sequence***, we mean

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \{x : \exists N_0 \in \mathbb{N}, \forall n > N_0, x \in A_n\}$$

2. By the ***Limit Superior of the sequence***,we mean

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \{x : \forall N \in \mathbb{N}, \exists n \geq N, x \in A_n\}$$

Note.

1. Limit Inferior 包含那些“最终稳定下来”的元素，即从某个点之后就永远属于序列中所有后续集合的元素。
2. Limit Superior 包含那些“反复出现”的元素，即在无限多个集合中出现的元素。

Proposition 1.4.2

For any sequence of sets $\{A_n\}$, it always holds that:

$$\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$$

Proof sketch.
直觉上是显然的，因为“最终稳定下来”的元素一定会“反复出现”.

Proof. It is obvious by the $\{x : x \in P\}$ form representation of the Limit Inferior and Superior. □

Definition 1.4.3 ▶ Limit of a Set Sequence

If the limit inferior and limit superior of a set sequence $\{A_n\}_{n=1}^{\infty}$ are equal, i.e., $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$, then we say the ***limit*** of the sequence exists, and it is defined as:

$$\lim_{n\to\infty} A_n = \liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$$

Proposition 1.4.4

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. If $\{A_n\}$ is an increasing sequence, then the limit of $\{A_n\}$ exists and

$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

2. If $\{A_n\}$ is a decreasing sequence, then the limit of $\{A_n\}$ exists and

$$\lim_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

Proof. 1. If $\{A_n\}$ is increasing , then

$$\bigcap_{n=N}^{\infty} A_n = A_N$$

and

$$\bigcup_{n=N}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n, \forall N \in \mathbb{N}.$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty} A_N$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

which are the same.

2. If $\{A_n\}$ is decreasing, then

$$\bigcap_{n=N}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\bigcup_{n=N}^{\infty} A_n = A_N$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} A_N$$

which are the same.

□

Proposition 1.4.5

For a sequence of sets $\{A_n\}_{n=1}^{\infty}$ and their corresponding sequence of indicator functions $\{\chi_{A_n}(x)\}_{n=1}^{\infty}$:

- $\chi_{\liminf_{n\to\infty} A_n}(x) = \liminf_{n\to\infty} \chi_{A_n}(x)$
- $\chi_{\limsup_{n\to\infty} A_n}(x) = \limsup_{n\to\infty} \chi_{A_n}(x)$
- $\lim_{n\to\infty} A_n$ exists, then $\chi_{\lim_{n\to\infty} A_n}(x) = \lim_{n\to\infty} \chi_{A_n}(x)$

Proof sketch.

- 若 x 在 limit inferior 里面, 则 x 是 "最终稳定" 的, $\chi_{A_n}(x)$ 是关于 n "最终" 恒为 1 的.
- 若 x 在 limit superior 里面, 则 x 是 "反复出现" 的, 即相当于 N 多大, 总会出现之后的某个 n 使得 $\chi_{A_n}(x) = 1$.
- 当极限存在时, 函数列 $\left\{\chi_{A_n}(x)\right\}_{n=1}^{\infty}$ 的 limit inferior 和 supperior 根据上两条相等, 等于极限集合的 χ.

1.5 Ordinal and Transfinite Induction

Definition 1.5.1 ▶ Ordinal

A set α is said to be an ***ordinal***, if the following two hold:

1. Transitive Set: If $x \in \alpha$, then $x \subseteq \alpha$. (Equivalently, if $z \in y$, $y \in \alpha$, then $z \in \alpha$).
2. Well-ordered by $\in$: For any $x, y \in \alpha$, either $x \in y$ or $y \in x$. And there is a $\in$-smallest element in any nonempty subset of α.

Lemma 1.5.2

If α is an ordinal, then $\alpha \notin \alpha$.

Proof. If $\alpha \in \alpha$, then $\{\alpha\}$ is a subset of α. But there is no smallest ordinal in $\{\alpha\}$, which is a contradiction. □

Definition 1.5.3 ▶ Classification of Ordinal

1. ***Zero Ordinal***: Only $0 = \varnothing$.
2. ***Successor Ordinal***: Ordinal numbers of the form $\alpha + 1 = \alpha \cup \{\alpha\}$
3. ***Limit Ordinal***: Ordinal that is neither zero ordinal nor successor ordinal.

Example 1.5.4 ▶ Ordinal

1. $$\omega = \{0, 1, 2, 3, \cdots\}$$
   is the smallest infinite ordinal, which is a limit ordinal.
2. $$\omega \cdot 2 := \{0, 1, 2, \cdots, \omega, \omega + 1, \omega + 2, \cdots\}$$

is a limit ordinal.

3. $$\omega_1 := \{\alpha : \alpha \text{ is a ountable ordinal}\}$$

is the smallest uncountable ordinal.

(a) ***Unountable***: If ω_1 is countable, then $\omega_1 \in \omega_1$ by definition, which is a contradiction.

(b) ***Smallest***: If there is a uncountable ordinal β such that $\beta < \omega_1$. Then $\beta \in \omega_1$, which is a contradiction to the uncountability of β.

Theorem 1.5.5 ▸ Transfinite Induction

For any ordinal α, let $P(\alpha)$ be a property of α. If the following holds :

- For any ordinal α, if $P(\beta)$ holds for any $\beta < \alpha$, then $P(\alpha)$ holds.

Then

- For any ordinal α, $P(\alpha)$ is true.

Proof. Let S be the collection of α such that $P(\alpha)$ is false. Then if the transfinite indunction is not true, S is nonempty. Since the class of ordinal is well-ordered, there is a smallest ordinal γ in S. Then for any $\beta < \gamma$, $\beta \notin S$, that is $P(\beta)$ is true. But it contradicts to what we supposed at first. □

1.6 Meager Sets in Topology

Definition 1.6.1 ▸ DEFINITIONS

- A subset E of a metric space X is said to be ***dense in an open set*** U if $\overline{E} \supset U$.
- A set E is defined to be ***nowhere dense*** if it is not dense in any open subset U of X. Alternatively, we could say that E is nowhere dense if $\overline{E}$ does not contain any open set.
- A set E is said to be of ***first category*** in X if it is the union of a countable collection of nowhere dense sets.
- A set that is not of the first category is said to be of the ***second category*** in X.

Theorem 1.6.2 ▶ Baire category theorem

A ***complete metric space*** X is not the union of a ***countable collection of nowhere dense sets***. That is, a ***complete metric space*** is of the ***second category***.